
Spring Boot

Testing Spring Boot Applications Demystified

A Hero's Journey Through the Spring Boot Testing Labyrinth

Talk @ Digiteal 23.06.2026

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Participate During the Talk

Go to menti.com and use the code **5463 9225** to **anonymously** submit answers for the quizzes and add your questions during the talk.

Start with the **first two questions**:

- Despite having LLMs and Code Agents, do you still write your tests by hand?
- Do You Enjoy Writing Automated Tests?

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Why Test Software?

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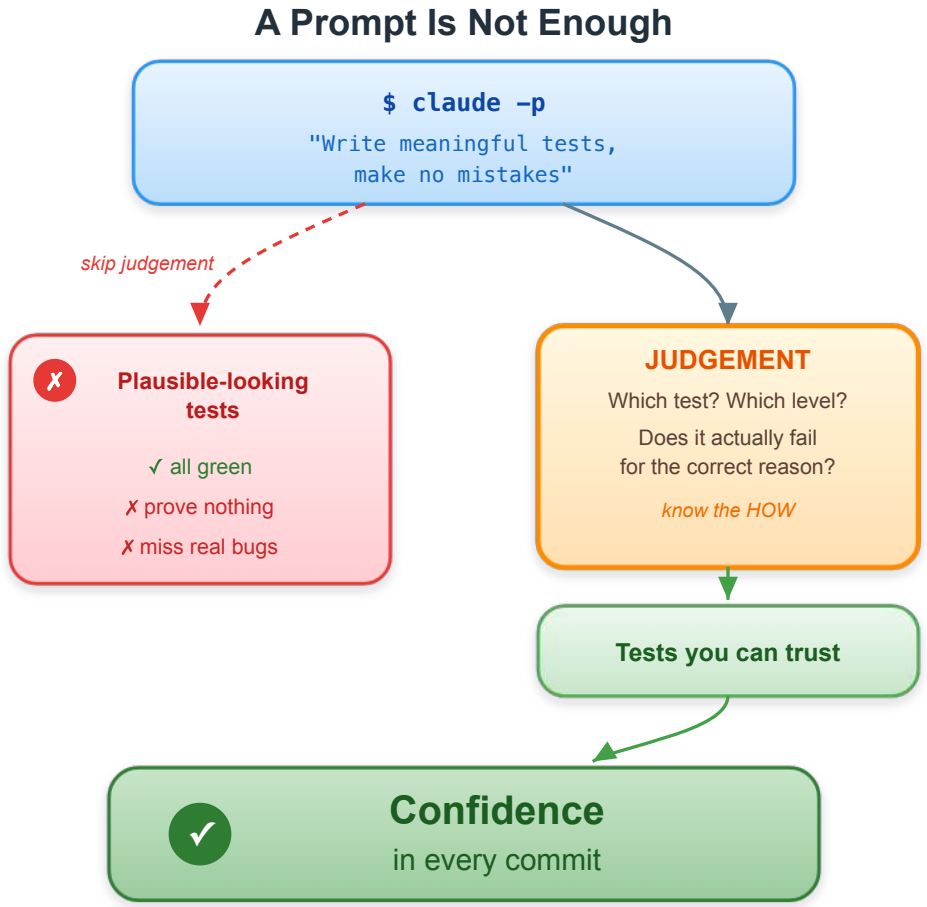
The AI Trap: Testing is Your Safety Net

- AI generates the logic, but you inherit the **liability**. It can write the function; it won't join the post-mortem.
- If the AI wrote the code and the AI wrote the test, you are the only person left to solve the **hallucination** when the **system crashes**.
- AI is the **accelerator**. Your tests are the **brakes**. You need both to go fast.

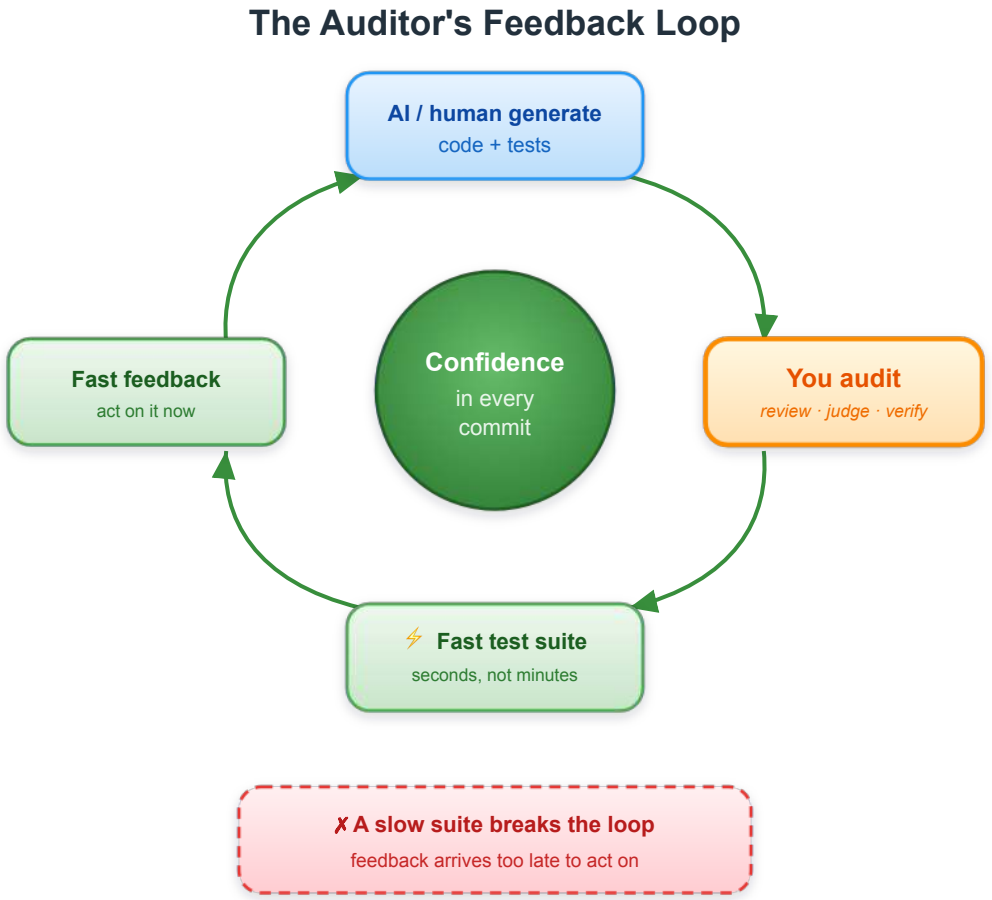
More Than a One-Line Prompt

- You can't just run `claude -p "Write meaningful tests, make no mistakes"` and walk away
- Effective AI testing needs **judgement** and understanding of the **HOW**: which test, at which level, and proof it actually fails
- The craft shifts from *writing* tests to *directing and reviewing* them - the goal stays **confidence in every commit**



From Developer to AI-Code Auditor

- AI writes more code, faster - your job becomes **auditing** it with confidence
- An auditor lives or dies by **fast feedback**: a slow suite breaks the loop
- Fast, reliable tests are the auditor's instrument - the foundation for **confidence in every commit**



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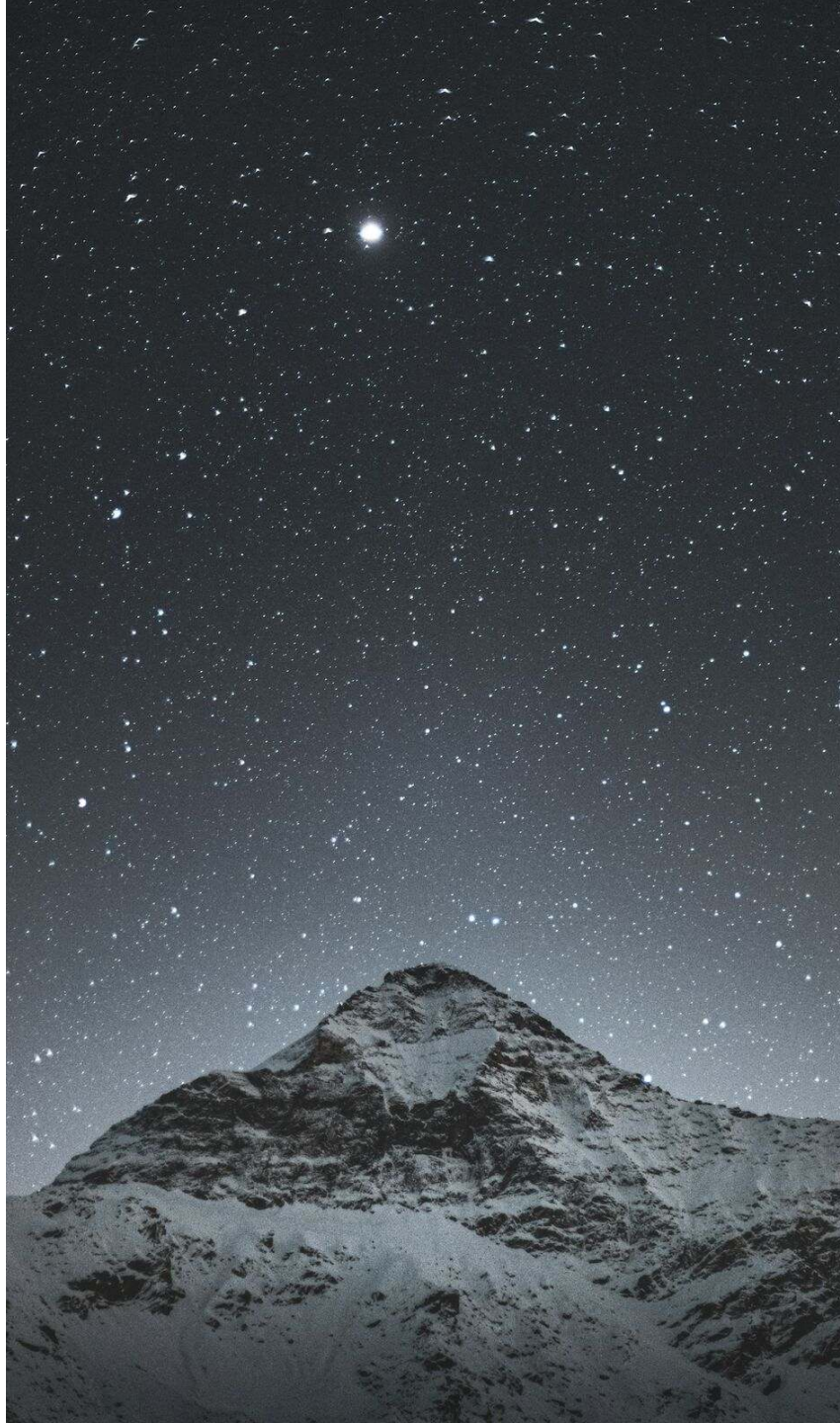
Good tests don't just catch bugs - they give you fast feedback and confident deployments.

My Overall Northstar for Automated Testing

Imagine seeing this pull request on a Friday afternoon:

How confident are you to merge this major Spring Boot upgrade and deploy it to production once the pipeline turns green?



The Hero's Journey aka. Our Agenda

Goals For This Talk

- **Provide a clear mental map** for choosing between unit, slice, and integration tests so developers stop guessing which tool to use
- **Equip attendees with practical techniques** to speed up test suites and validate test quality
- **Build confidence to ship fearlessly** by creating tests that catch real bugs, not just achieve coverage metrics

About Philip

- Self-employed developer from Herzogenaurach, Germany (Bavaria) 🍻
- Blogging & content creation with a focus on testing Java and specifically Spring Boot applications 🍃
- Founder of PragmaTech GmbH - Enabling Developers to Frequently Deliver Software with More Confidence 🚤
- Enjoys writing tests (sometimes even more than production code) 🧪

Act 1: The Grand Entrance

Testing Spring Boot applications can feel like entering a labyrinth blindfolded:

- Copying test configuration from AI/StackOverflow hoping it works
- The paradox of choice: `@SpringBootTest`, `@WebMvcTest`, `@DataJpaTest`, `@MockBean`, etc.
- Struggling with Spring `ApplicationContext` creation during tests
- Uncertainty about what to test and how to test it effectively

Quest 1

The Unit Testing Guardian

The Swift Gatekeeper - Blocks Those Who Overcomplicate

Our Foundation: Spring Boot Starter Test

- The "Testing Swiss Army Knife"

```xml
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-test</artifactId>
  <scope>test</scope>
</dependency>
```

- Batteries-included for testing by transitively including popular testing libraries
- Out-of-the-box dependency management to ensure compatibility

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```
./mvnw dependency:tree
[INFO] ...
[INFO] +- org.springframework.boot:spring-boot-starter-test:jar:4.0.2:test
[INFO] |  +- org.springframework.boot:spring-boot-test:jar:4.0.2:test
[INFO] |  +- org.springframework.boot:spring-boot-test-autoconfigure:jar:4.0.2:test
[INFO] |  +- com.jayway.jsonpath:json-path:jar:2.10.0:test
[INFO] |  |  \- org.slf4j:slf4j-api:jar:2.0.17:compile
[INFO] |  +- jakarta.xml.bind:jakarta.xml.bind-api:jar:4.0.4:test
[INFO] |  |  \- jakarta.activation:jakarta.activation-api:jar:2.1.4:test
[INFO] |  +- net.minidev:json-smart:jar:2.6.0:test
[INFO] |  |  \- net.minidev:accessors-smart:jar:2.6.0:test
[INFO] |  |     \- org.ow2.asm:asm:jar:9.7.1:test
[INFO] |  +- org.assertj:assertj-core:jar:3.27.6:test
[INFO] |  |  \- net.bytebuddy:byte-buddy:jar:1.17.8:test
[INFO] |  +- org.awaitility:awaitility:jar:4.3.0:test
[INFO] |  +- org.hamcrest:hamcrest:jar:3.0:test
[INFO] |  +- org.junit.jupiter:junit-jupiter:jar:6.0.2:test
[INFO] |  |  +- org.junit.jupiter:junit-jupiter-api:jar:6.0.2:test
[INFO] |  |  |  +- org.opentest4j:opentest4j:jar:1.3.0:test
[INFO] |  |  |  +- org.junit.platform:junit-platform-commons:jar:6.0.2:test
[INFO] |  |  |  \- org.apiguardian:apiguardian-api:jar:1.1.2:test
[INFO] |  |  +- org.junit.jupiter:junit-jupiter-params:jar:6.0.2:test
[INFO] |  |  \- org.junit.jupiter:junit-jupiter-engine:jar:6.0.2:test
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[INFO] |  +- org.mockito:mockito-core:jar:5.5.0:test
[INFO] |  |  +- net.bytebuddy:byte-buddy-agent:jar:1.17.8:test
[INFO] |  |  \- org.objenesis:objenesis:jar:3.3:test
[INFO] |  +- org.mockito:mockito-junit-jupiter:jar:5.5.0:test
[INFO] |  +- org.skyscreamer:jsonassert:jar:1.5.3:test
[INFO] |  |  \- com.vaadin.external.google:android-json:jar:0.0.20131108.vaadin1:test
[INFO] |  +- org.springframework:spring-core:jar:7.0.3:compile
[INFO] |  |  +- commons-logging:commons-logging:jar:1.3.5:compile
[INFO] |  |  \- org.jspecify:jspecify:jar:1.0.0:compile
[INFO] |  +- org.springframework:spring-test:jar:7.0.3:test
[INFO] |  \- org.xmlunit:xmlunit-core:jar:2.10.4:test
```

JUnit: The Testing Foundation

- Java's de-facto standard testing framework - version **6** with Spring Boot 4 (drop-in upgrade, unlike 4 → 5)
- More than just `@Test`: **Extension API** (replaces `@RunWith`), lifecycle hooks, `@ParameterizedTest`, `@Nested`, `@DisplayName`, **Parallel execution**

```
1 @ExtendWith(MockitoExtension.class)
2 class DiscountCalculatorTest {
3
4   @ParameterizedTest
5   @CsvSource({ "0, 0.0", "100, 10.0", "1000, 150.0" })
6   void shouldApplyDiscountWhenAmountIsValid(int amount, double expected) {
7     // ...
8   }
9 }
```

Mockito: Stub, Verify, and Beyond

- **Stubbing** with `when(...).thenReturn(...)` and **verifying** interactions with `verify(...)`
- **Argument matchers** (`any()`, `eq()`, `argThat(...)`) for flexible expectations
- **Advanced**: deep stubs (`RETURNS_DEEP_STUBS`) and static mocking (`MockedStatic`) for legacy/utility code

```
1 when(customerRepository.findById(42L))
2   .thenReturn(Optional.of(customer));
3
4 verify(eventPublisher).publish(any(CustomerCreated.class));
```

AssertJ & Hamcrest: Readable Assertions

- **AssertJ**: fluent, chainable, IDE-friendly auto-completion
- **Hamcrest**: composable matchers, useful with matcher-driven APIs like Mockito `argThat(...)`
- Pick one for general assertions and stick with it within a test class

```
1 // AssertJ – chainable & expressive
2 assertThat(customers)
3   .hasSize(3)
4   .extracting(Customer::firstName)
5   .containsExactly("Alice", "Bob", "Charlie");
6
7 // Hamcrest – composable matchers
8 assertThat("duke".toUpperCase(), equalTo("DUKE"));
```

JsonPath & JSONAssert: Working With JSON

- **JsonPath**: query JSON like XPath - drill into responses without deserializing
- **JSONAssert**: compare JSON strings with **lenient** or **strict** mode - order-insensitive, ignores extra fields

```
 1 String json = "{ ... }";
 2
 3 // JsonPath – query a JSON document directly
 4 String firstName = JsonPath.parse(json).read("$.firstName", String.class);
 5 Long tagCount = JsonPath.parse(json).read("$.tags.length()", Long.class);
 6
 7 // JSONAssert – lenient: extra fields in actual are OK
 8 String expected = "{ \"name\": \"duke\" }";
 9 String actual = "{ \"name\": \"duke\", \"age\": 42 }";
10
11 JSONAssert.assertEquals(expected, actual, JSONCompareMode.LENIENT);
```

XMLUnit: Comparing XML Documents

- Compare and validate XML with **whitespace-aware**, **namespace-aware**, **order-tolerant** diffs
- Still relevant for SOAP, configuration files, and legacy enterprise integrations

```
String control = "<customer>...</customer>";
String test = "<customer>...</customer>";

Diff diff = DiffBuilder.compare(control)
  .withTest(test)
  .ignoreWhitespace()
  .checkForSimilar()
  .build();

assertThat(diff.hasDifferences()).isFalse();
```

Awaitility: Taming Asynchronous Tests

- Fluent **polling** for eventually-consistent assertions - message queues, async event handlers, scheduled jobs
- Replaces brittle `Thread.sleep(...)` with explicit conditions and timeouts

```
1 await()
2   .atMost(5, SECONDS)
3   .pollInterval(100, MILLISECONDS)
4   .untilAsserted(() ->
5     assertThat(orderRepository.findAll()).hasSize(1)
6   );
```

Unit Testing Java/Spring Boot Applications 101

- **Core Concept**: Test individual components in isolation from dependencies - one unit of work at a time.
- **Confidence Gained**: Fast, high-volume verification that the smallest building blocks behave correctly under various conditions.
- **Pitfall**: Poor class design leads to untestable god classes. Good tests start with good design.
- **Tools**: JUnit, Mockito, AssertJ (or Spock, TestNG, Hamcrest).

Unit Testing has Limits

Consider this sample REST controller, what could we verify with a unit test?

```java
@RestController
@RequestMapping("/api/customers")
public class CustomerController {

  private final CustomerService customerService;

  public CustomerController(CustomerService customerService) {
    this.customerService = customerService;
  }

  @PostMapping
  public ResponseEntity<Void> createNewCustomer(@Validated CustomerCreationRequest payload, UriComponentsBuilder uriBuilder) {

    String customerId = customerService.createNewCustomer(payload.firstName());

    UriComponents uriComponents = uriBuilder
      .path("/api/customers/{id}")
      .buildAndExpand(customerId);

    return ResponseEntity.created(uriComponents.toUri()).build();
  }
}
```

```java
@ExtendWith(MockitoExtension.class)
class CustomerControllerUnitTests {

  @Mock
  private CustomerService customerService;

  @InjectMocks
  private CustomerController customerController;

  @Test
  void shouldCreateCustomerWhenPayloadRequestIsValid() {
    when(customerService.createNewCustomer(anyString()))
      .thenReturn("42");

    ResponseEntity<Void> result = customerController.createNewCustomer(
      new CustomerCreationRequest("Java", "Duke", "duke@spring.io"),
      UriComponentsBuilder.newInstance()
    );

    assertThat(result.getStatusCode().value())
      .isEqualTo(201);
    assertThat(result.getHeaders().getLocation().toString())
      .isEqualTo("/api/customers/42");
  }
}
```

Things We Can't Cover With a Unit Test #0

Request Mapping: Does HTTP GET `/api/customers/{id}` actually resolve to our desired method?

```
@Test
void shouldCreateCustomerWhenPayloadRequestIsValid() {

  // ...

  ResponseEntity<Void> result = customerController.createNewCustomer(
    new CustomerCreationRequest("Java", "Duke", "duke@spring.io"),
    UriComponentsBuilder.newInstance()
  );
}
```

Things We Can't Cover With a Unit Test #1

Validation: Will an incomplete request body result in a 400 bad request or return an accidental 201?

```
 1 @Test
 2 void shouldCreateCustomerWhenPayloadRequestIsValid() {
 3
 4   // ...
 5
 6   ResponseEntity<Void> result = customerController.createNewCustomer(
 7     new CustomerCreationRequest("Java", "Duke", "NOT_AN_EMAIL"),
 8     UriComponentsBuilder.newInstance()
 9   );
10 }
```

Things We Can't Cover With a Unit Test #2

Serialization: Are we JSON objects serialized and deserialized correctly?

```
1 ResponseEntity<Void> result = customerController.createNewCustomer(
2   new CustomerCreationRequest("Java", "Duke", "NOT_AN_EMAIL"),
3   UriComponentsBuilder.newInstance());
```

```
1 {
2   "first-name": "Java",
3   "last_Name": "Duke",
4   "email": "duke@spring.io"
5 }
```

Things We Can't Cover With a Unit Test #3

Security: Are we Spring Security configuration and other authorization checks enforced?

```
@Test
void shouldCreateCustomerWhenPayloadRequestIsValid() {

  // ...

  ResponseEntity<Void> result = customerController.createNewCustomer(
    new CustomerCreationRequest("Java", "Duke", "NOT_AN_EMAIL"),
    UriComponentsBuilder.newInstance()
  );
}
```

Quest 2

The Slice Testing Hydra

Multiple Heads, Each Guarding a Layer

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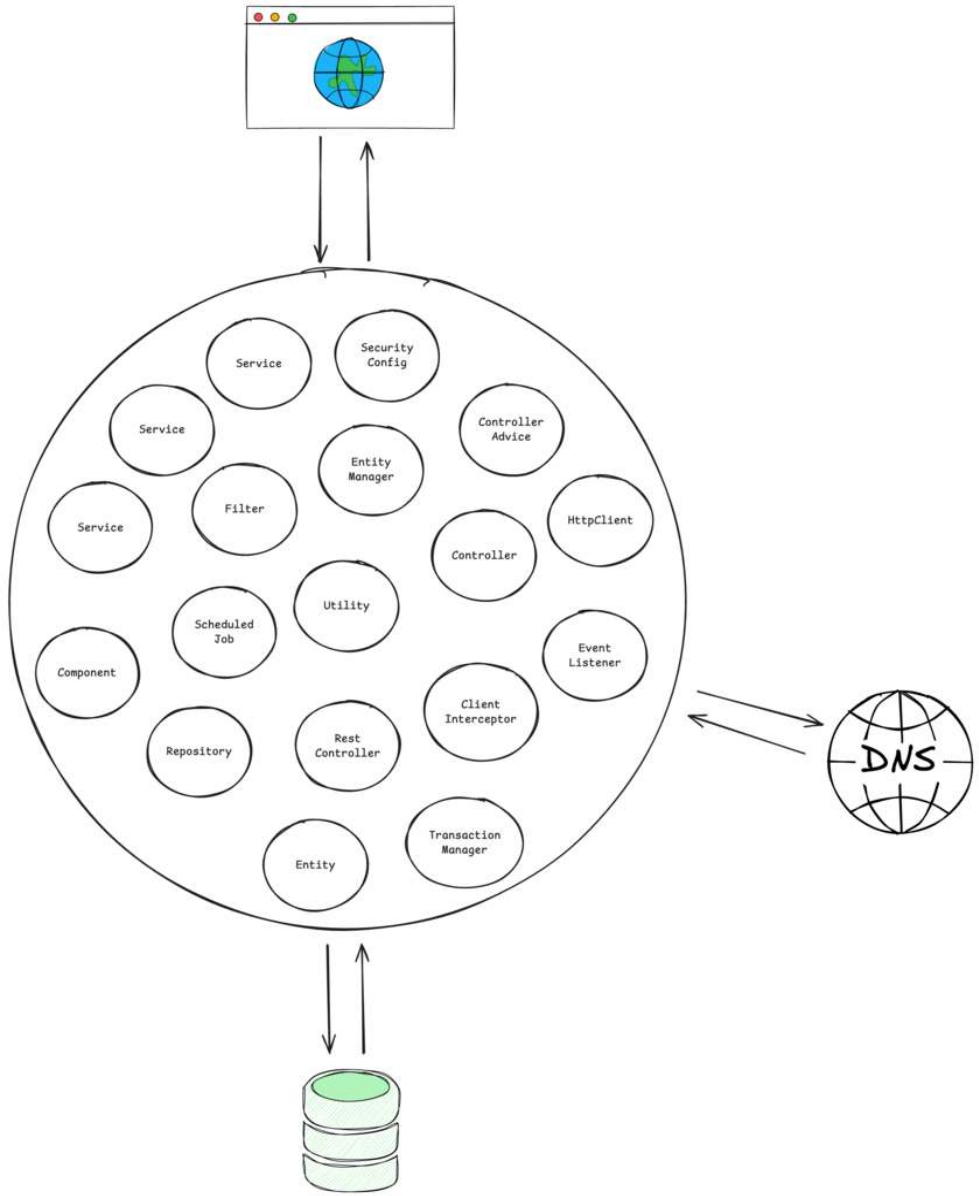
A Typical
Spring ApplicationContext
with different
Spring Beans



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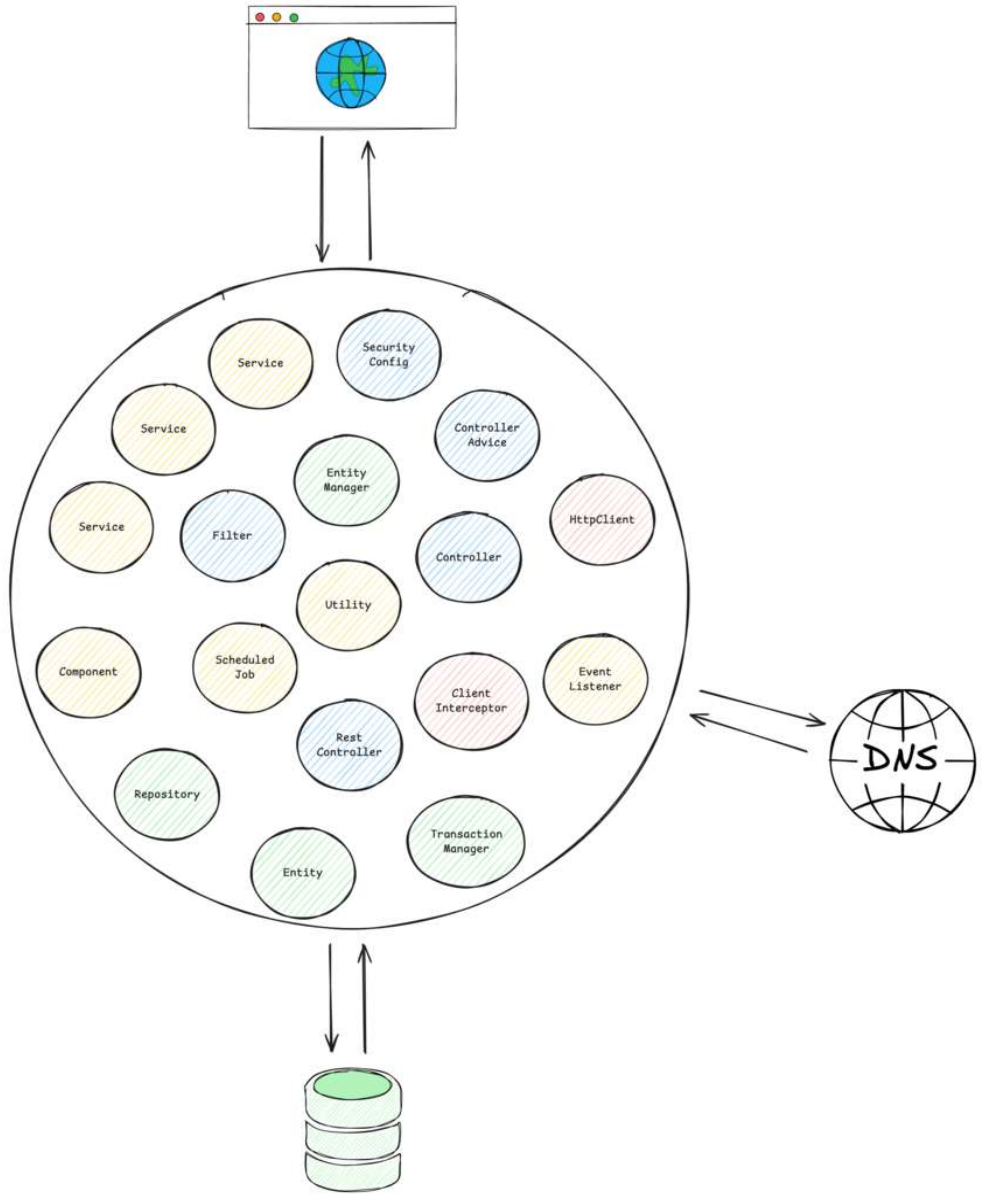
A Typical
Spring ApplicationContext



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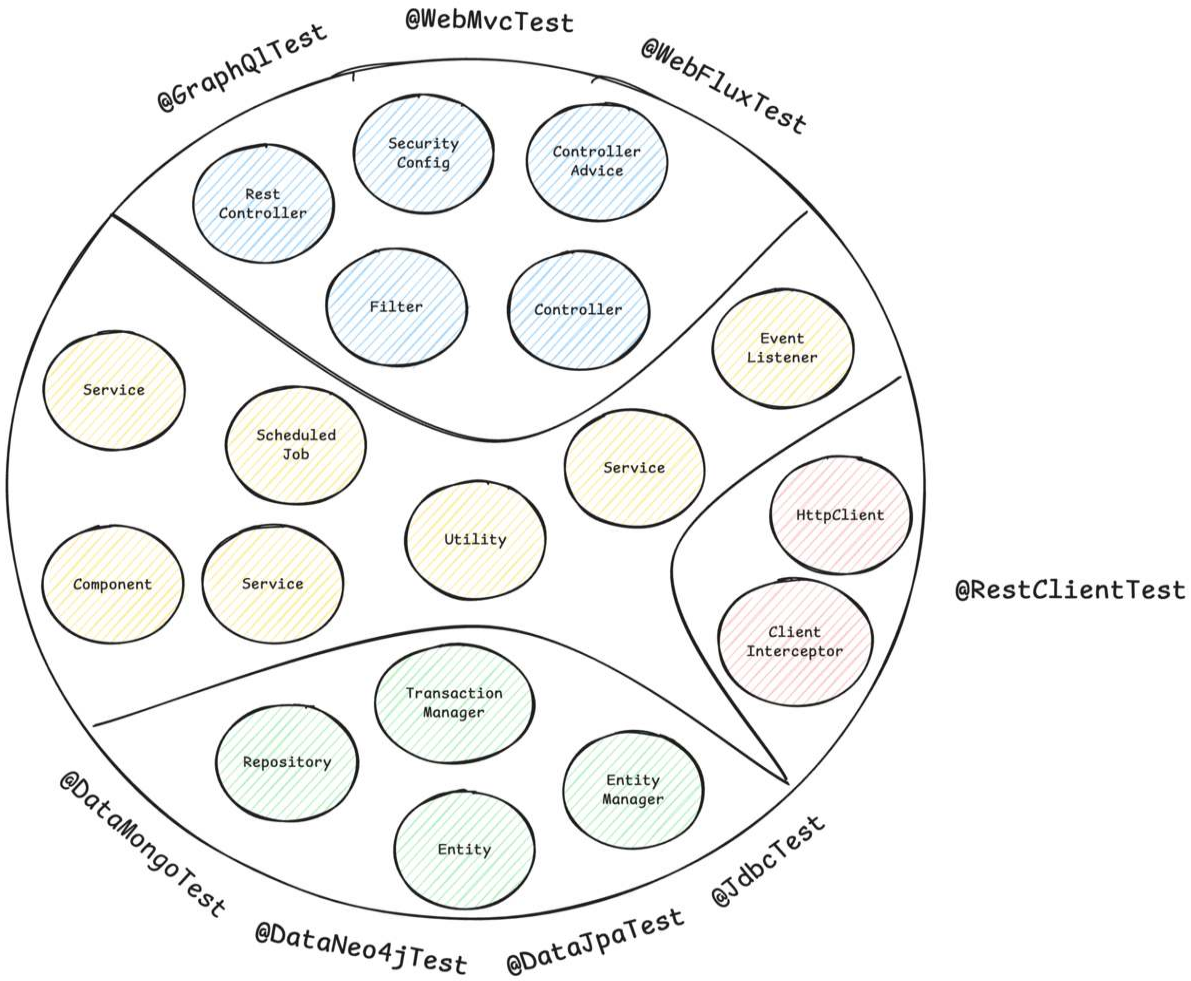

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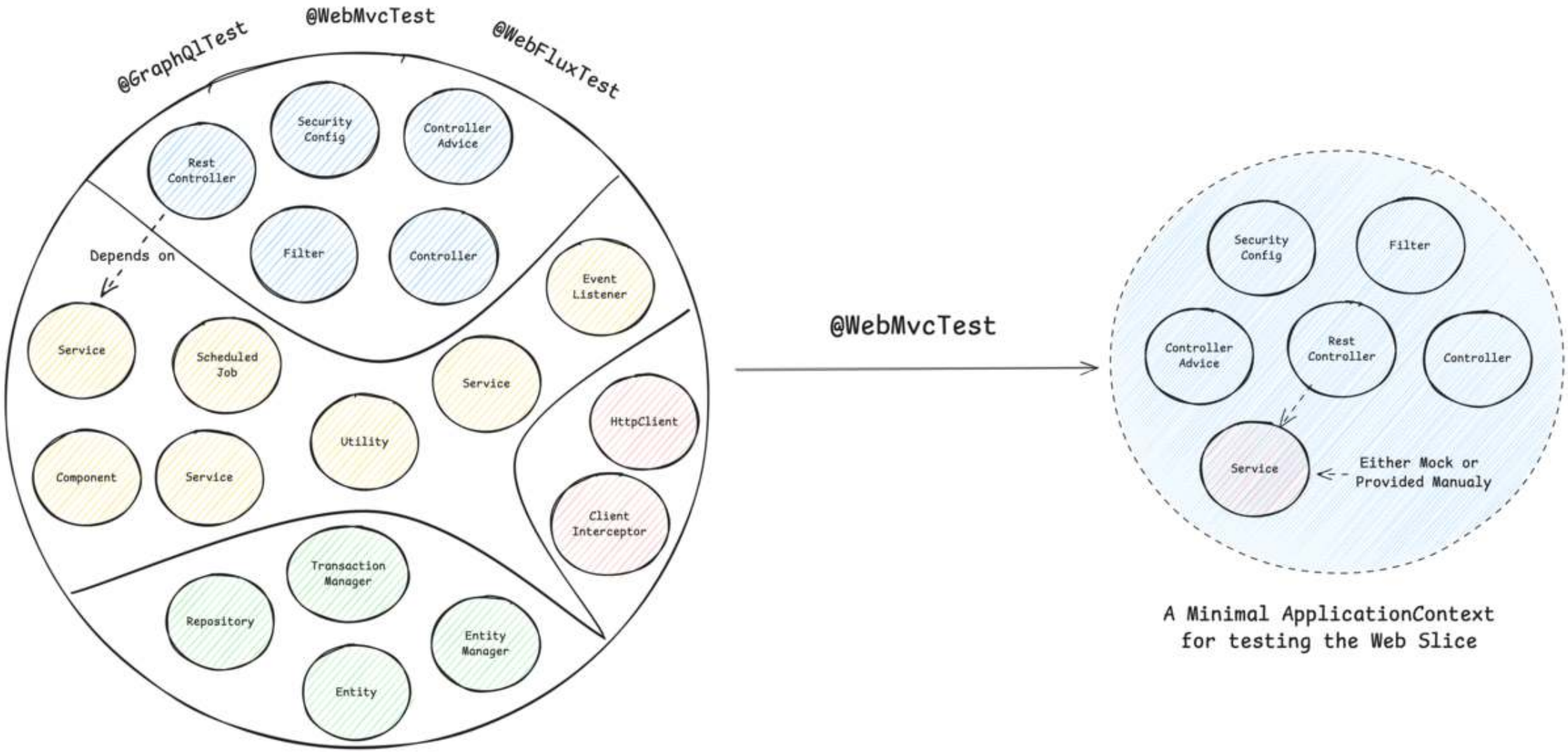

Spring Boot Test Slice Example: `@WebMvcTest`

```java
@WebMvcTest(CustomerController.class)
@Import(SecurityConfig.class)
class CustomerControllerTest {

  @Autowired
  private MockMvc mockMvc;

  @MockitoBean
  private CustomerService customerService;

  @Test
  @WithMockUser
  void shouldReturnLocationOfNewlyCreatedCustomer() throws Exception {
    // ...
  }
}
```

Common Test Slices

- `@WebMvcTest` / `@WebFluxTest` - Controller layer
- `@DataJpaTest` / `@JdbcTest` - Persistence layer
- `@JsonTest` - JSON serialization/deserialization
- `@RestClientTest` - RestTemplate testing
- etc.

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@DataCouchbaseTest

@DataLdapTest

@RestClientTest

@JsonTest

@JooqTest

@WebFluxTest

@DataRedisTest

@WebMvcTest

@DataMongoTest

@DataCassandraTest

@GraphQlTest

@DataNeo4jTest

@WriteYourOwn*

@DataJpaTest

@SqsTest

@JdbcTest

@DataElasticsearchTest

Sliced Testing Spring Boot Applications 101

- **Core Concept**: Test a specific "slice" or layer of your application by loading a minimal, relevant part of the Spring `ApplicationContext`.
- **Confidence Gained**: Helps validate parts of your application where pure unit testing is insufficient, like the web, messaging, or data layer.
- **Prominent Examples:** Web layer (`@WebMvcTest`) and database layer (`@DataJpaTest`)
- **Pitfalls**: Requires careful configuration to ensure only the necessary slice of the context is loaded.
- **Tools**: JUnit, Mockito, Spring Test, Spring Boot, Testcontainers

Quest 3

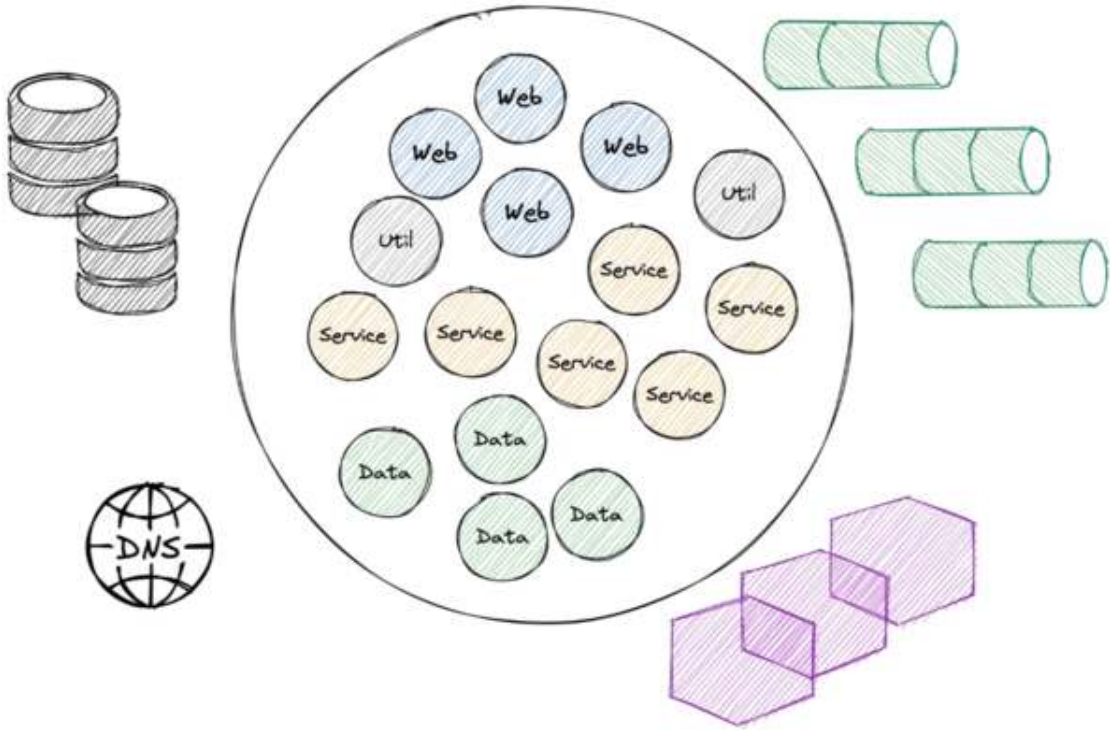
The Integration Testing Dragon

Guards the Full Treasure - but Demands Patience

```java
@SpringBootTest
class ApplicationTest {

  @Test
  void contextLoads() {
  }

}
```



Challenges when Starting the Entire `ApplicationContext`

- **Problem #1**: How to Ensure Surrounding Infrastructure (e.g. database, queues, etc.) is Present?
- **Problem #2**: How to Interact with our Application for Integration Tests?
- **Problem #3**: How to Keep our Build Time at a reasonable Duration?

There's Even More...

- **Problem #4**: How to handle HTTP communication from our application to remote services?
- **Problem #5**: How to provide test data and maintain a clean state between tests?
- **Problem #6**: How to handle authentication and security contexts during tests?

Provide External Infrastructure with Testcontainers (Problem #1)

Running infrastructure components (databases, message brokers, etc.) in Docker containers for our tests becomes a breeze with Testcontainers:

```
1 @Container
2 @ServiceConnection
3 static PostgreSQLContainer<?> postgres = new PostgreSQLContainer<>("postgres:16-alpine")
4   .withDatabaseName("testdb")
5   .withUsername("test")
6   .withPassword("test")
7   .withInitScript("init-postgres.sql");
```

This gives us an ephemeral PostgreSQL database for our tests:

```
1 $ docker ps
2 CONTAINER ID   IMAGE                         COMMAND                  CREATED          STATUS          PORTS                                                NAMES
3 a958ee2887c6   postgres:16-alpine            "docker-entrypoint.s…"   10 seconds ago   Up 9 seconds    0.0.0.0:32776->5432/tcp, [::]:32776->5432/tcp        affectionate_cannon
4 ad0f804068dc   testcontainers/ryuk:0.12.0    "/bin/ryuk"              10 seconds ago   Up 9 seconds    0.0.0.0:32775->8080/tcp, [::]:32775->8080/tcp        testcontainers-ryuk-1f9f76a6-46d4-4e19-85c1-e8364da12804
```

How to Interact with our Application for Integration Tests?

Option #1:

```java
@SpringBootTest
// which is @SpringBootTest(webEnvironment = SpringBootTest.WebEnvironment.MOCK)
@AutoConfigureMockMvc
class ApplicationMockWebIT {

  // @LocalServerPort
  // private int port; <-- this would fail the test, there is no local port occupied

  @Test
  @WithMockUser
  void givenCustomersThenReturnListForAuthenticatedUser(@Autowired MockMvc mockMvc) throws Exception {
    mockMvc
      .perform(get("/api/customers")
        .header(ACCEPT, APPLICATION_JSON))
      .andExpect(status().is(200))
      .andExpect(content().contentType(APPLICATION_JSON))
      .andExpect(jsonPath("$.size()", is(1)));
  }
}
```

How to Interact with our Application for Integration Tests?

Option #2:

```java
@AutoConfigureWebTestClient // required since Spring Boot 4
@SpringBootTest(webEnvironment = SpringBootTest.WebEnvironment.RANDOM_PORT)
class ApplicationServletContainerIT {

  @Autowired
  WebTestClient webTestClient;

  @Test
  void contextLoads() {
    webTestClient
      .get()
      .uri("/api/customers")
      .header("Authorization", "Basic " + Base64.getEncoder().encodeToString("user:dummy".getBytes()))
      .exchange()
      .expectStatus()
      .isOk();
  }
}
```

Integration Testing Spring Boot Applications 101

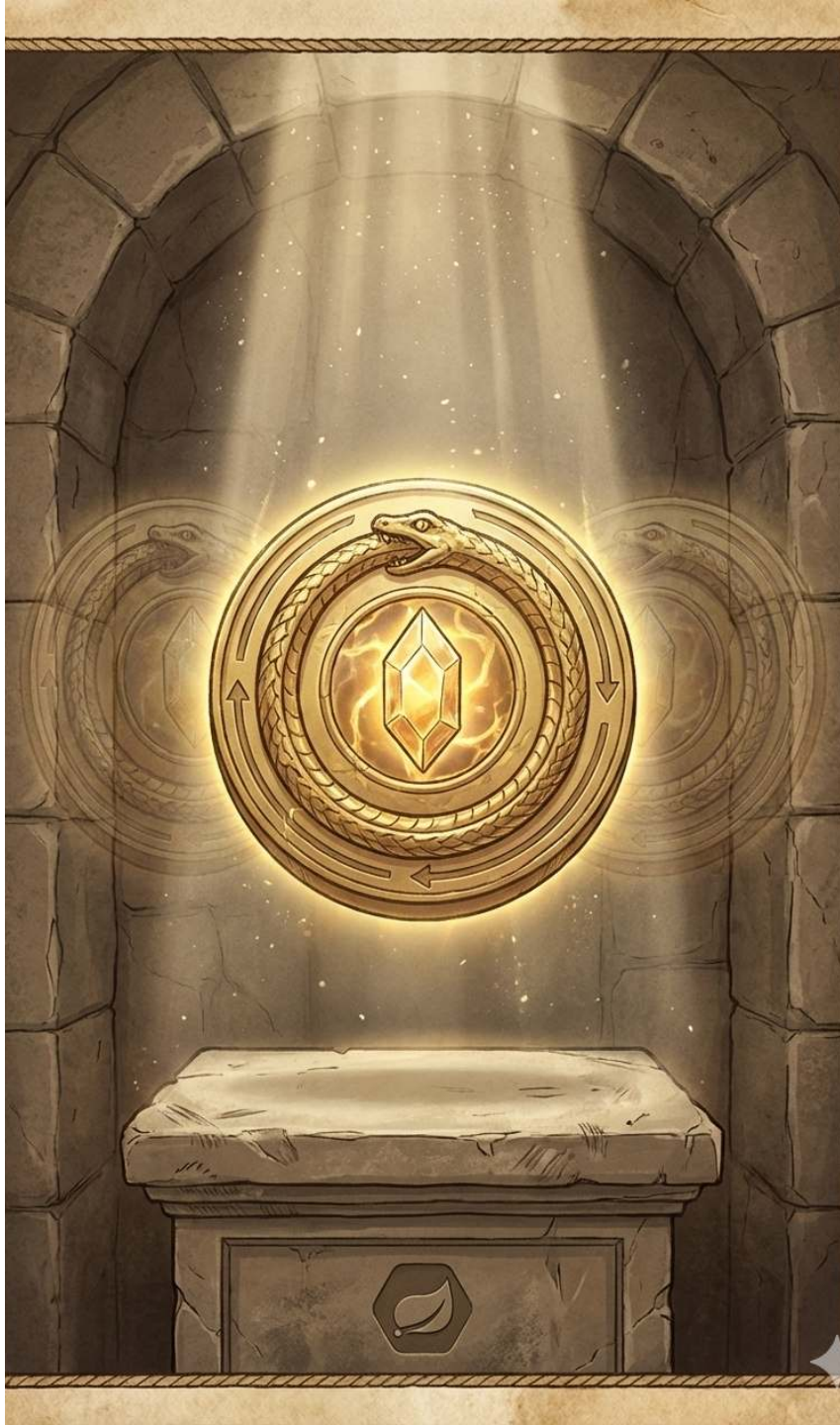
- **Core Concept**: Start the entire Spring application context, often on a random local port, and test the application through its external interfaces (e.g., REST API).
- **Confidence Gained**: Validates the integration of all internal components working together as a complete application.
- **Best Practices**: Use `@SpringBootTest` to run the app on a local port.
- **Pitfalls**: Slower to run than unit or sliced tests. Managing the lifecycle of dependent services can be complex.
- **Tools**: JUnit, Mockito, Spring Test, Spring Boot, Testcontainers, WireMock (for mocking external HTTP services), Selenium (for browser-based UI testing)

… but what about **Problem #3**: How to Keep our Build Time at a reasonable Duration?

Quest Item 1

The Caching Amulet

Helps You Reuse What You Already Built



Integration Testing - The Need for Speed

- **The Problem:** Integration tests require a started & initialized Spring `ApplicationContext`, which slows down the build
- **The Solution:** Spring Test `TestContext` Caching – stores an already started Spring `ApplicationContext` for reuse
- This feature is part of Spring Test (included in every Spring Boot project via `spring-boot-starter-test`)

Example of speed improvement:

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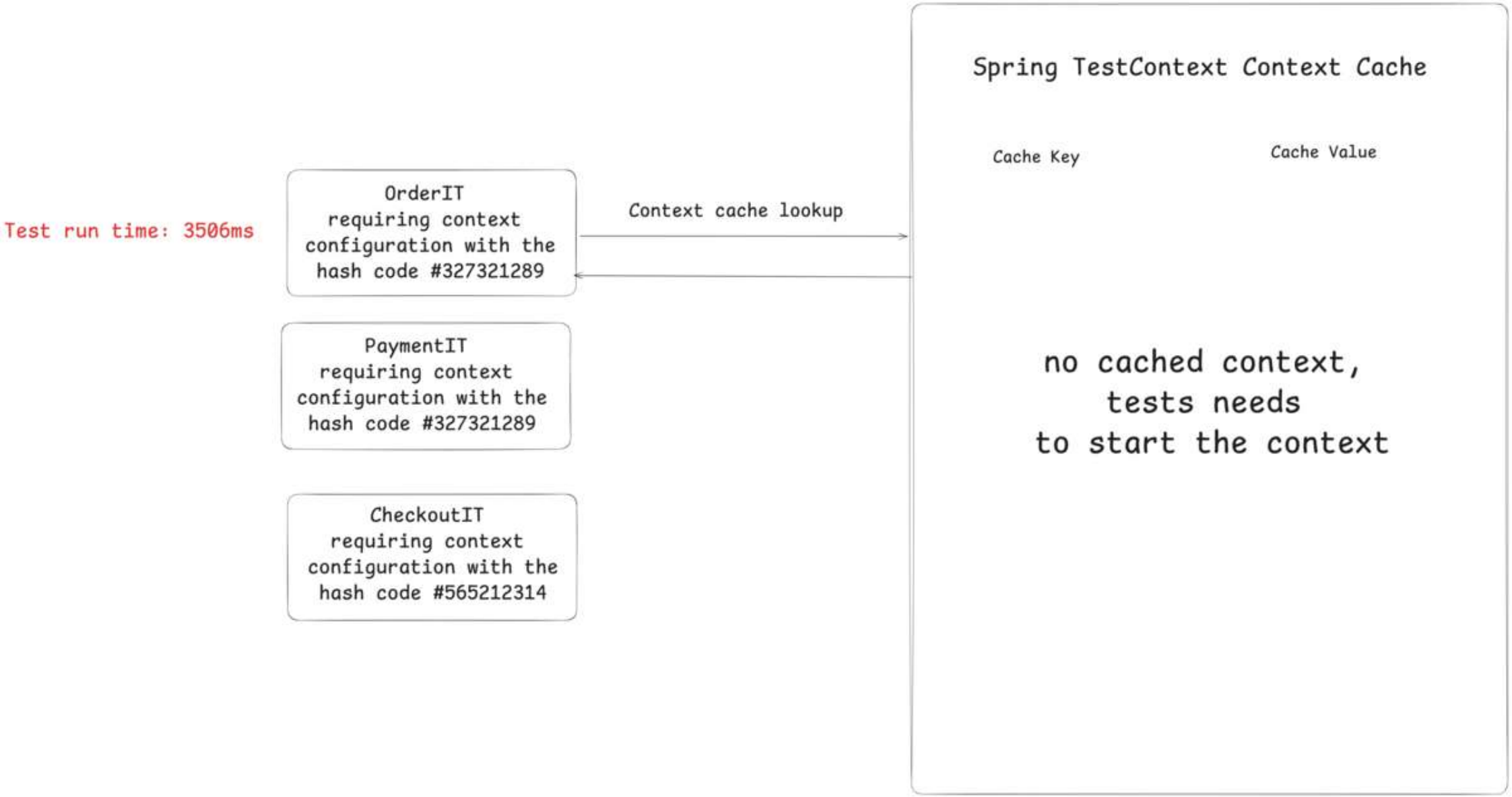
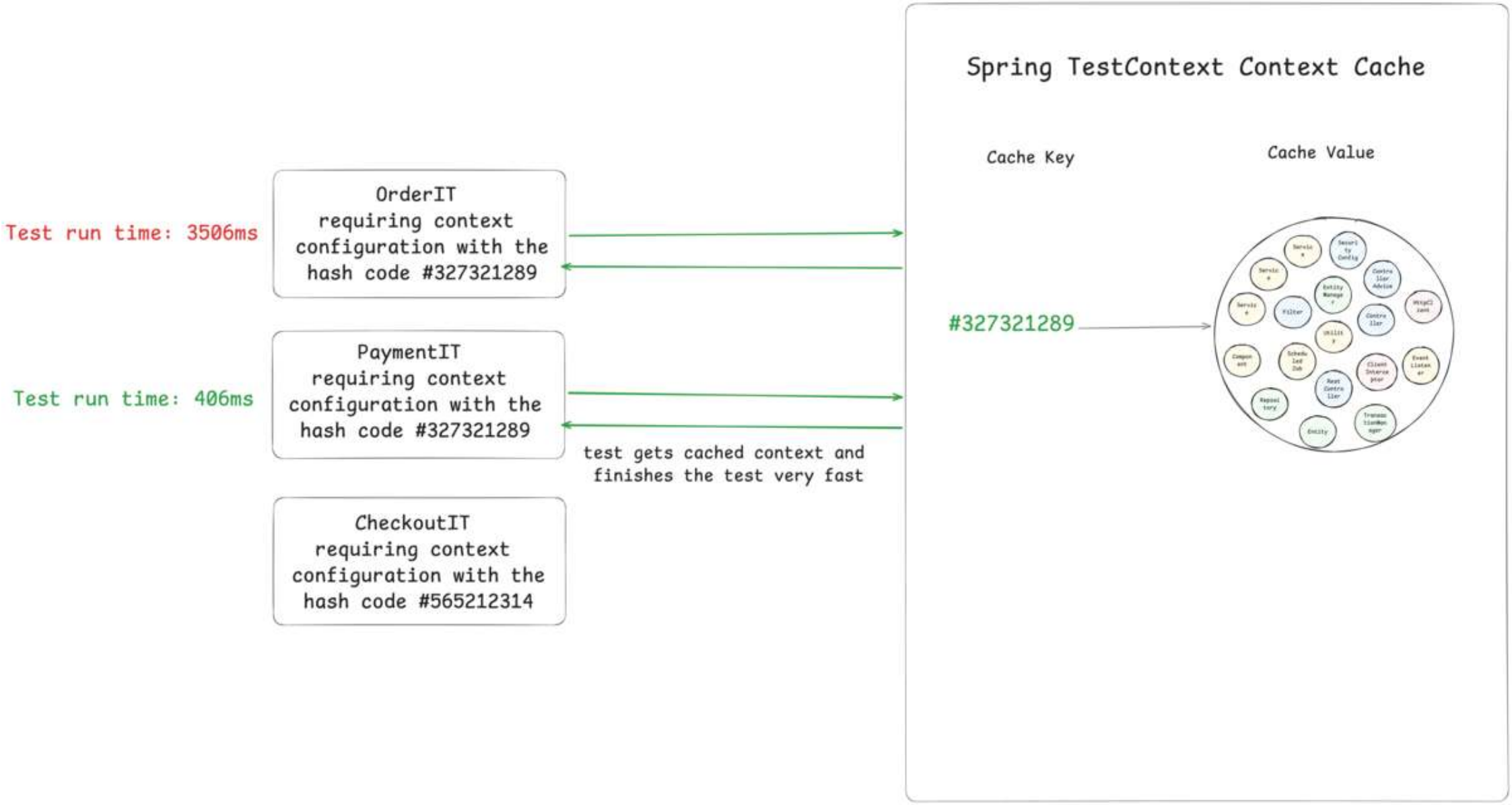
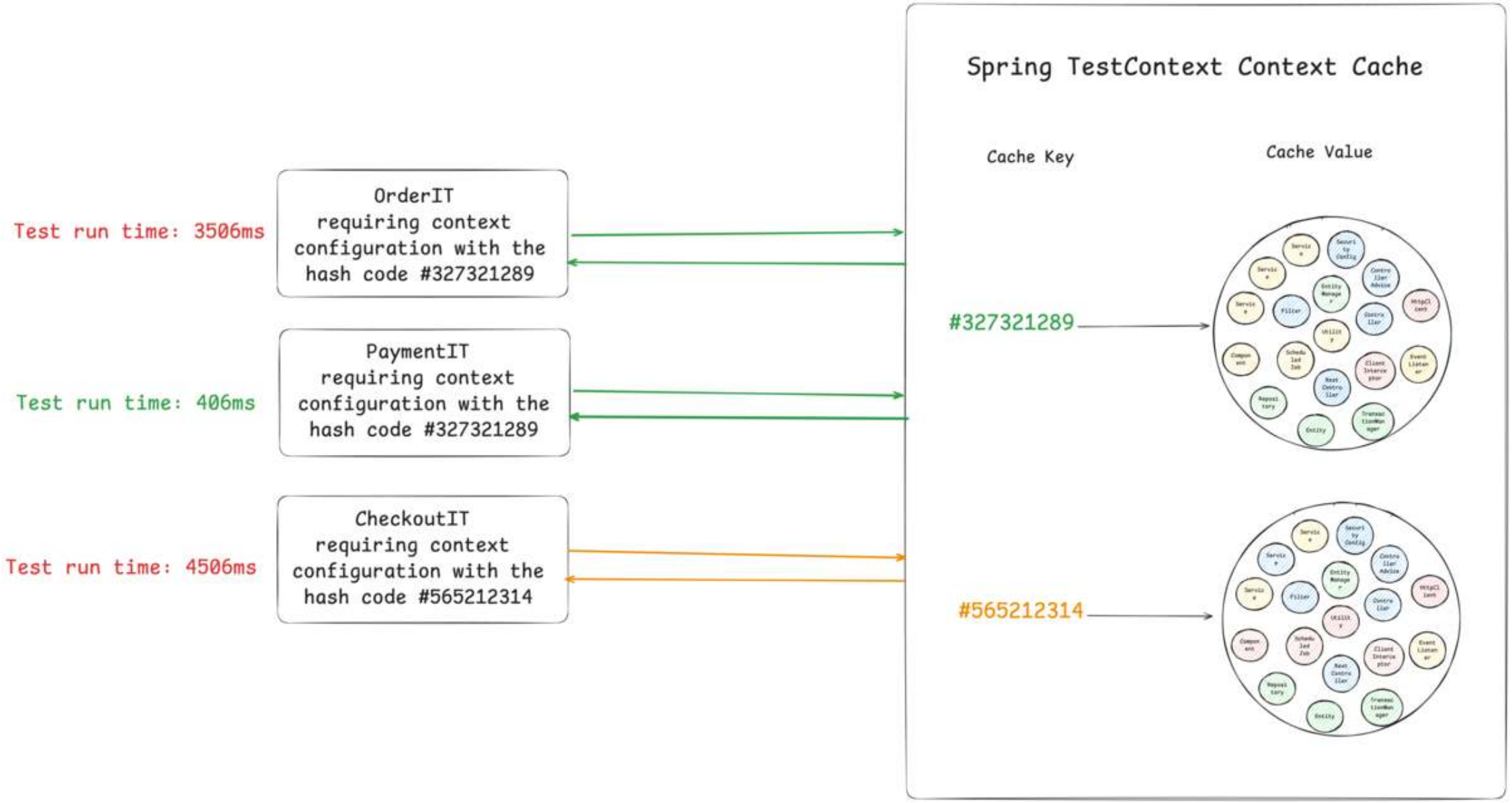



Spring TestContext Context Cache
Cache Key
Cache Value
#327321289
Test run time: 3506ms
OrderIT
requiring context
configuration with the
hash code #327321289
Test run time: 406ms
PaymentIT
requiring context
configuration with the
hash code #327321289
test gets cached context and
finishes the test very fast
CheckoutIT
requiring context
configuration with the
hash code #565212314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hash code #327321289
#327321289
Test run time: 406ms
PaymentIT
requiring context
configuration with the
hash code #327321289
Test run time: 4506ms
CheckoutIT
requiring context
configuration with the
hash code #565212314
#565212314

How the Cache Key is Built

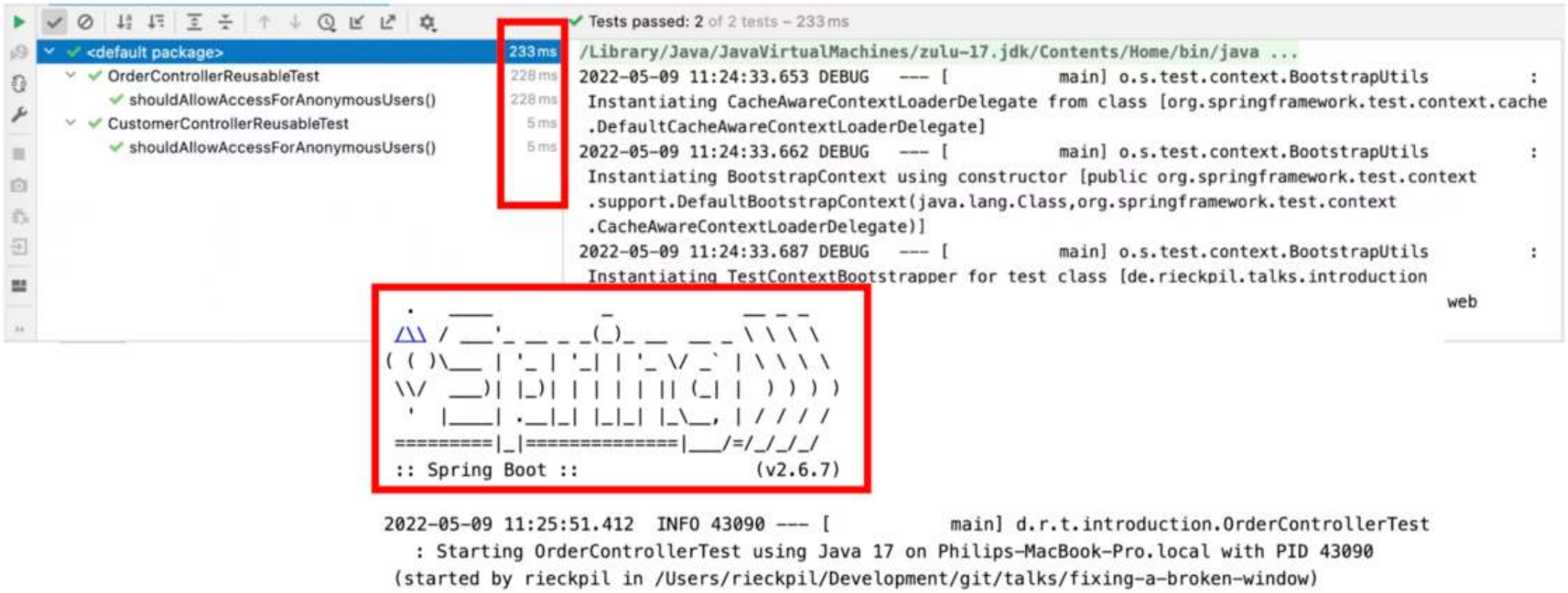
```
1 // DefaultContextCache.java
2 private final Map<MergedContextConfiguration, ApplicationContext> contextMap =
3   Collections.synchronizedMap(new LinkedHashMap<>(32, 0.75f, true));
```

The following information is part of the Cache Key (`MergedContextConfiguration`):

- activeProfiles (`@ActiveProfiles`)
- contextInitializersClasses (`@ContextConfiguration`)
- propertySourceLocations (`@TestPropertySource`)
- propertySourceProperties (`@TestPropertySource`)
- contextCustomizer (`@MockitoBean` , `@MockBean` , `@DynamicPropertySource` , ...)
- etc.

```
Test class
    │
    ▼
MergedContextConfiguration(
  testClass, locations, classes,
  activeProfiles, propertyValues,
  contextInitializers, contextCustomizers   ← every @MockitoBean lands here
  ... etc.
)
    │
    ▼  hashCode() / equals()

Cache hit? → reuse context ✅
Cache miss? → start new context and store it 🆕
```

Detect Context Restarts - Visually



Detect Context Restarts - with Logs

```
<logger name="org.springframework.test.context" level="TRACE" />
```

```
2022-05-05 14:27:38.136 DEBUG 42500 --- [           main] org.springframework.test.context.cache   : Spring test
 ApplicationContext cache statistics: [DefaultContextCache@68e62b3b size = 1, maxSize = 32, parentContextCount = 0,
 hitCount = 11, missCount = 1]
2022-05-05 14:27:38.136 DEBUG 42500 --- [           main] tractDirtiesContextTestExecutionListener : After test class:
 context [DefaultTestContext@325bb9a6 testClass = ApplicationIT, testInstance = [null], testMethod = [null], testException
  = [null], mergedContextConfiguration = [WebMergedContextConfiguration@1d12b024 testClass = ApplicationIT, locations =
 '{}', classes = '{class de.rieckpil.talks.Application}', contextInitializerClasses = '[]', activeProfiles = '{}',
 propertySourceLocations = '{}', propertySourceProperties = '{org.springframework.boot.test.context
 .SpringBootTestContextBootstrapper=true, server.port=0}', contextCustomizers = set[org.springframework.boot.test.context
 .filter.ExcludeFilterContextCustomizer@5215cd9a, org.springframework.boot.test.json
 .DuplicateJsonObjectContextCustomizerFactory$DuplicateJsonObjectContextCustomizer@9257031, org.springframework.boot.test
```

Detect Context Restarts - with Tooling

An open-source Spring Test utility that provides visualization and insights for Spring Test execution, with a focus on Spring context caching statistics.

Overall goal: Identify optimization opportunities in your Spring Test suite to speed up your builds and ship to production faster and with more confidence.

The Final Boss

Developers tend to consult AI/StackOverflow for integration test issues and often copy advice from the internet without knowing the implications:

```
1 @SpringBootTest
2 @DirtiesContext
3 // this instructs Spring to remove the context from the cache
4 // and rebuild a new context on every request
5 public abstract class AbstractIntegrationTest {
6
7 }
```

The setup above will **disable** the context caching feature and slow down the builds significantly!

New in Spring Framework 7: Pausing Contexts

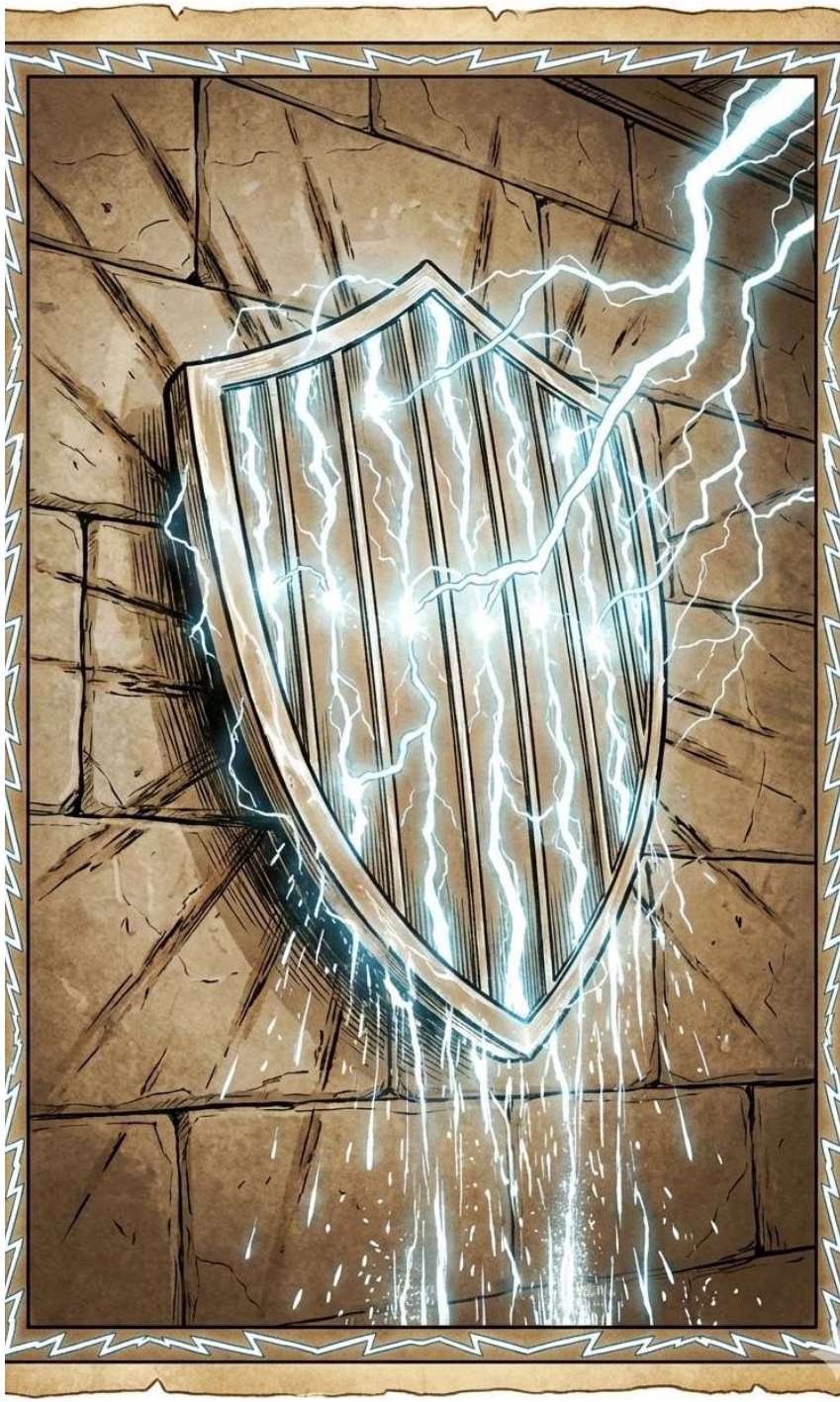
See Release Notes von Spring Framework 7.

> Pausing of Test Application Contexts
>
> The Spring TestContext framework is caching application context instances within test suites for faster runs. As of Spring Framework 7.0, we now pause test application contexts when
> they're not used.
>
> This means an application context stored in the context cache will be stopped when it is no longer actively in use and automatically restarted the next time the context is retrieved from the cache.
>
> Specifically, the latter will restart all auto-startup beans in the application context, effectively restoring the lifecycle state.

Quest Item 2

The Lightning Shield

Many Cores, One Goal



Test Parallelization

Goal: Reduce build time and get faster feedback

Requirements:

- No shared state
- No dependency between tests and their execution order
- No mutation of global state

Two ways to achieve this:

- Fork a new JVM with Surefire/Failsafe (or for the Gradle test task) and let it run in parallel
- Use JUnit Jupiter's parallelization mode and let it run in the same JVM with multiple threads

Java Test Parallelization Options

Maven Surefire Fork-Based

(Multiple JVMs)

JVM Fork 1

- Test Class A
- Test Class B
- Test Class C

JVM Fork 2

- Test Class D
- Test Class E
- Test Class F

Maven (forkCount) Features:

- Each fork is a separate JVM process
- Isolated memory spaces
- Higher memory overhead
- High isolation (one DB per fork)
- Configuration: forkCount, reuseForks

JUnit Jupiter Thread-Based

(Single JVM)

Single JVM

Thread Pool

Thread 1	Thread 2
Test Method A1	Test Method A2
Test Method B1	Test Method B2
Test Class C	Test Class D

JUnit Jupiter Features:

- Uses threads within a single JVM
- Shared memory space
- Lower memory overhead
- Configuration: junit.jupiter.execution.parallel.*

Test Parallelization 101

Using Surefire/Failsafe:

```
<plugin>
  <artifactId>maven-surefire-plugin</artifactId>
  <configuration>
    <forkCount>1C</forkCount> <!-- 1 JVM per CPU core -->
  </configuration>
</plugin>
```

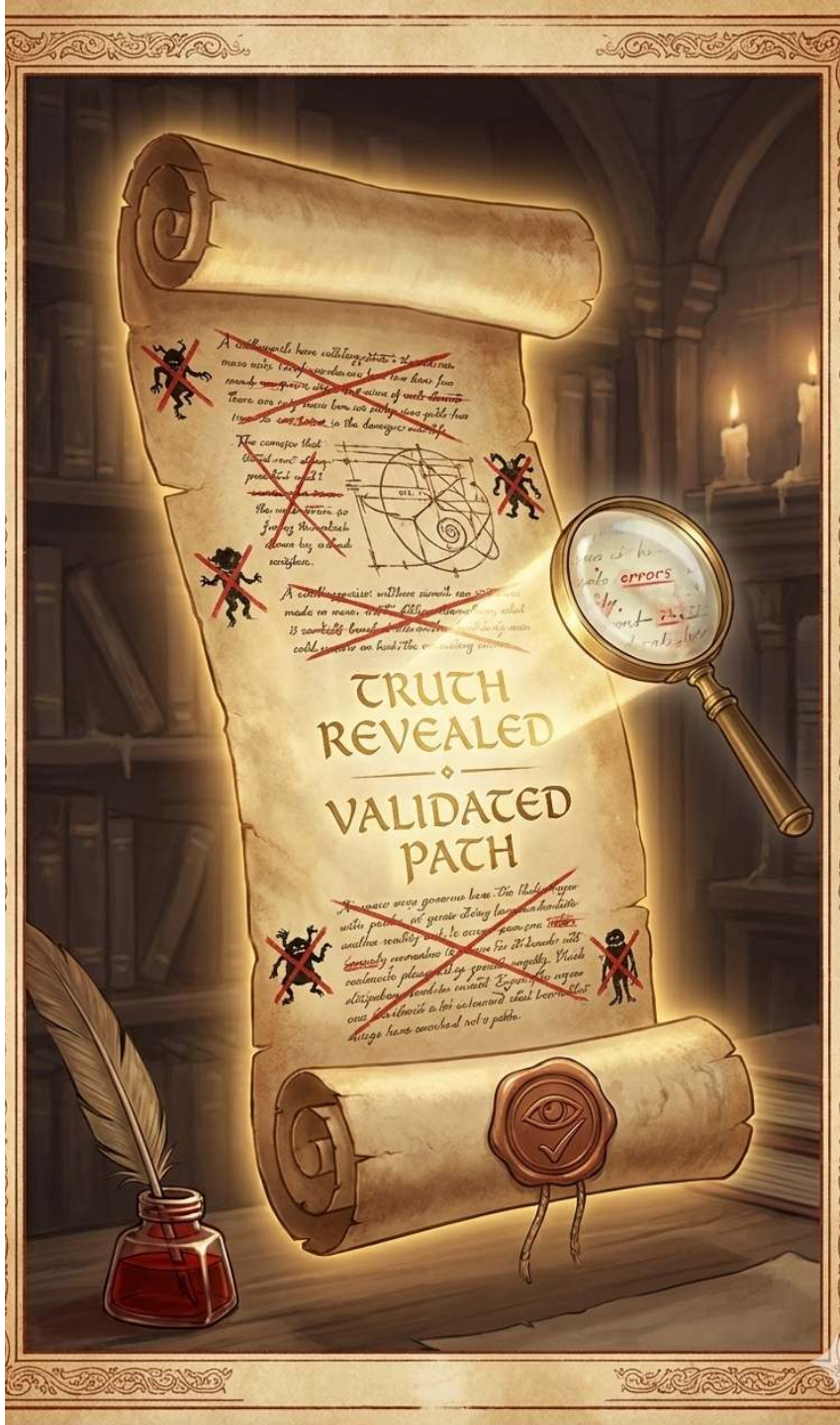
With JUnit Jupiter:

```
# src/test/resources/junit-platform.properties
junit.jupiter.execution.parallel.enabled = true
junit.jupiter.execution.parallel.mode.default = same_thread
junit.jupiter.execution.parallel.mode.classes.default = concurrent
```

Quest Item 3

The Scroll of Truth

Coverage Lies, Mutants Don't

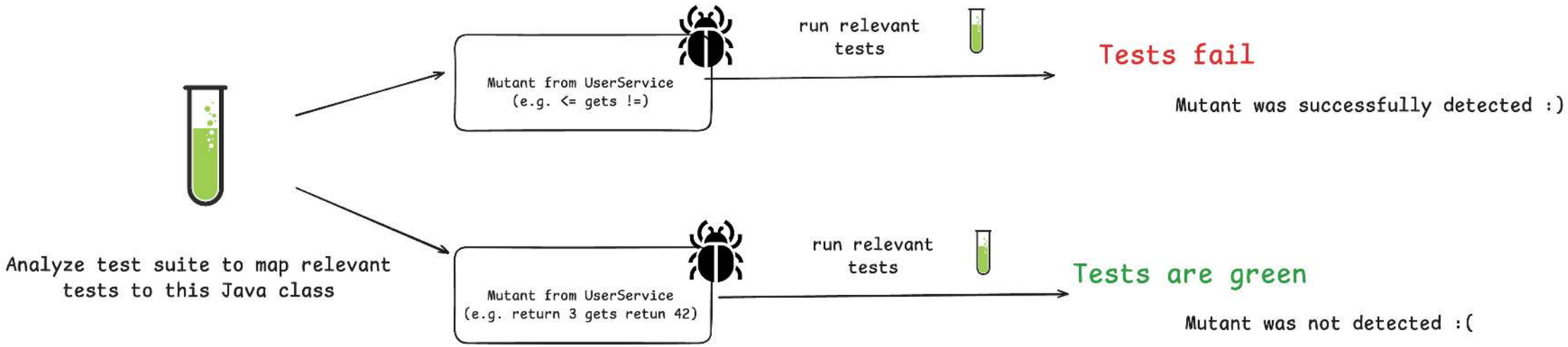


Let's Challenge Code Coverage

```java
public Long registerUser(int age, String username) {

  if (age <= 18) {
    throw new IllegalArgumentException("User must be at least 18 years old");
  }

  if ("ADMIN".equalsIgnoreCase(username)) {
    throw new IllegalArgumentException("Username 'ADMIN' is not allowed");
  }

  // ...

}
```

Idea: Introduce Regressions to Verify Test Quality

```java
if (age != 18) { // PIT mutated this with bytecode manipulation
    throw new IllegalArgumentException("User must be at least 18 years old");
}
```




```java
if ("ADMIN".equals(username)) {
    return 42L; // PIT mutated this with bytecode manipulation
}
```

Introducing: Mutation Testing

- Having high code coverage might give you a **false sense of security**
- Mutation Testing with PIT
- Beyond Line Coverage: Traditional tools like JaCoCo show which code runs during tests, but PIT verifies if our tests actually detect when code behaves incorrectly by introducing "**mutations**" to our source code.
- Quality Guarantee: PIT automatically **modifies our code** (changing conditionals, return values, etc.) to ensure our tests fail when they should, **revealing blind spots** in seemingly comprehensive test suites.

Act 5: The Triumphant Exit

- Spring Boot applications come with batteries-included and excellent testing support
- We've completed three main quests: Unit testing, Sliced testing, and Integration testing
- Three core quest items help us to speed up and validate our tests:
  - Context Caching
  - Test Parallelization
  - Mutation Testing

Spring Boot 4 - Testing Support Keeps Improving

- **RestTestClient**: Modern, fluent alternative for the `TestRestTemplate` / `WebTestClient` / `RestAssured`.
- **Context pausing**: Cached test contexts are now automatically paused, eliminating resource conflicts from background processes.
- **JUnit 6**: Drop-in upgrade from JUnit 5 - far smoother than the JUnit 4 → 5 migration.
- **Testcontainers 2.0**: New `testcontainers-` prefix for modules, JUnit 4 support removed.
- **Bean overrides for non-singletons**: `@MockitoBean` and `@TestBean` now work with prototype and custom-scoped beans.

Upcoming Open Online Workshops

Confidence In Every Commit: Essentials (1 Day) - Achieve confidence in every commit. Stop fighting your test suite and start mastering it. Covering fast & reliable unit, sliced and integration testing with Spring Boot

Next dates:

- 🗓️ **Thursday 02.07.2026**: 9 AM - 4 PM CEST
- 🗓️ **Tuesday 08.09.2026**: 9 AM - 4 PM CEST

See the detailed agenda and save your spot.

What we Cover in the Workshop

- Section 1: Foundations & The Testing Pyramid 2.0: Moving beyond "coverage" to "confidence."
- Section 2: Sliced Contexts & Unit Testing
- Section 3: Real-World Integration Testing
- Section 4: Performance & Strategy
- Section 5: Live Q&A & Implementation

Let's Tailor Your Team's Agentic Workflow

A **hands-on workshop** to develop and establish your **unified agentic guardrails** for **optimized testing and confidence in every commit**.

```
spring-boot-testing/
├── SKILL.md                          # router: picks the cheapest test level, links to subskills
├── unit-testing/
│   ├── SKILL.md                      # JUnit 5 + Mockito, no Spring context
│   └── examples.md
├── slice-testing/
│   ├── SKILL.md                      # @DataJpaTest, @RestClientTest, @JsonTest, MockWebServer
│   ├── examples.md
│   └── reference.md                  # annotation surface + Boot 4 import paths
├── slice-testing-webmvc/
│   ├── SKILL.md                      # @WebMvcTest + MockMvc + security
│   └── examples.md
├── integration-testing/
│   ├── SKILL.md                      # @SpringBootTest + shared base class
│   └── examples.md
├── testcontainers-setup/
│   ├── SKILL.md                      # @ServiceConnection vs DynamicPropertySource, singletons/reuse
│   ├── examples.md
│   └── reference.md                  # @ServiceConnection support matrix, lifecycle cheat-sheet
└── test-setup-reviewer/
    ├── SKILL.md                      # audit workflow (context-cache + parallel safety)
    ├── reference.md                  # anti-pattern catalogue + JUnit parallel config
    └── scripts/
        └── audit_test_setup.py       # working scanner (tested on this repo)
```

Talk @ Digiteal 23.06.2026 - Questions @ menti.com Code: **5463 9225**

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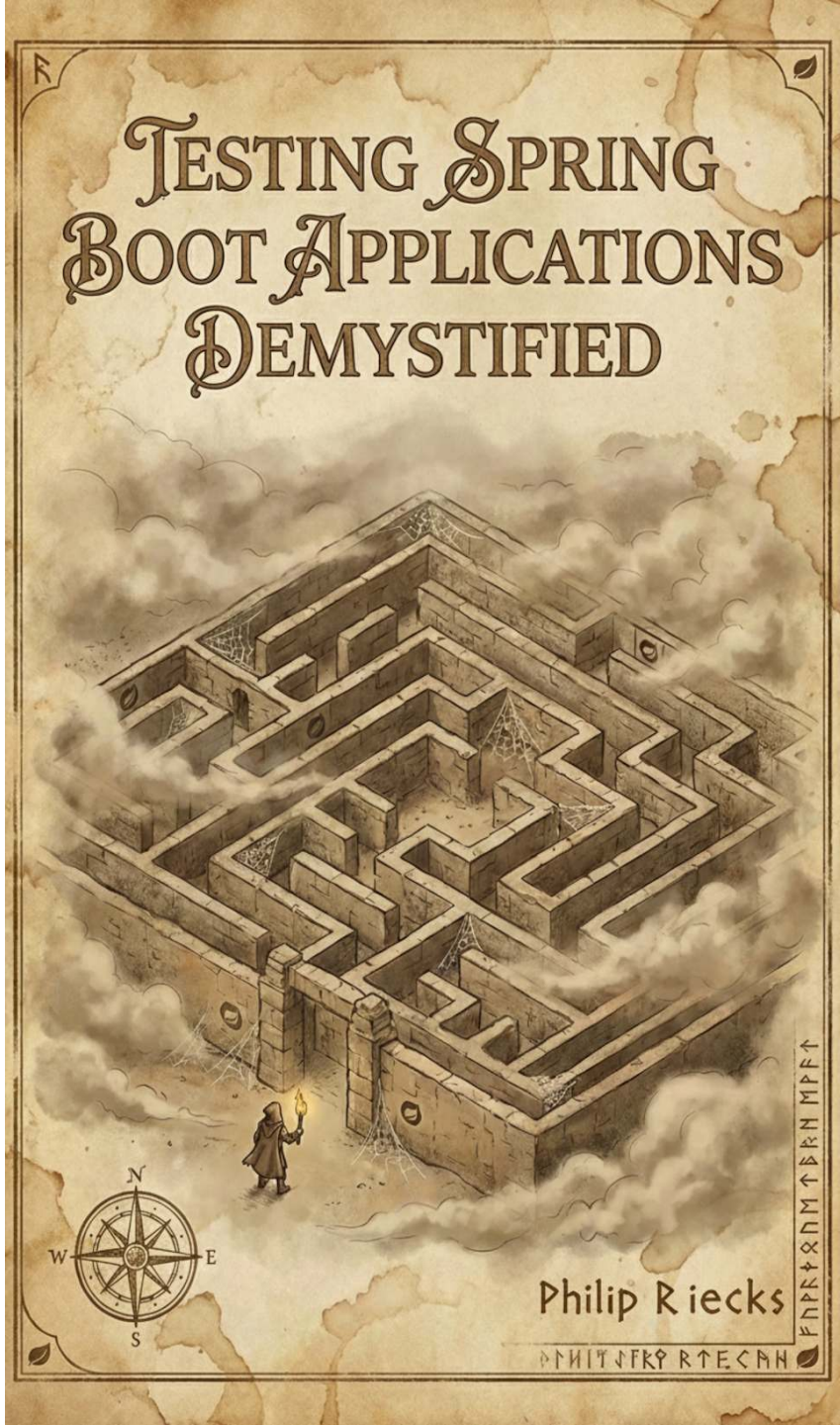

Joyful Testing!

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Reach out any time via: LinkedIn (Philip Riecks) or Mail (philip@pragmatech.digital)